Multi-Plan Telecommunications Cost Optimization

Complete worked solution

Each plan has a fixed price, included data, voice minutes, validity days, and an excess rate per GB and minute. A plan is eligible when validity covers the requested days, its included resources meet minimum required resources, and its estimated total fits the budget. Choose the lowest total among eligible plans; if tied, prefer more included data, then shorter plan name alphabetically.

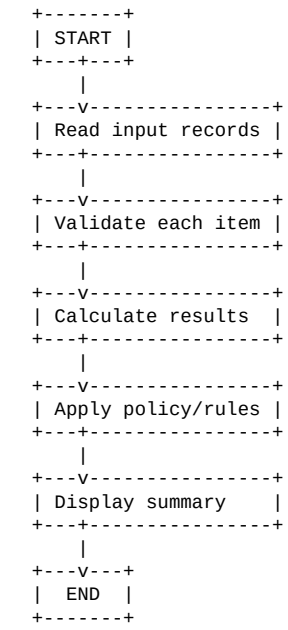
1. Computational Thinking and Algorithm

Inputs: Supplied records and policy values. **Processing:** Validate data, calculate charges or scores, apply ordered rules, and compare outcomes. **Outputs:** Itemized results, classifications, rankings, or recommendations.

Pseudocode

```
START
  Read usage, days, budget, and plan list
  FOR each plan
    Validate plan values
    Compute excess data and minutes above included amounts
    Add excess charges to fixed price
    IF validity, minimum resources, and budget are satisfied
      Add (cost, -included data, plan name) to candidates
  IF candidates exist, sort and display best plan
  ELSE report that no plan meets all constraints
END
```

2. Flowchart



3. Python Program

The program uses basic variables, numeric and string data, tuples, lists, loops, conditionals, arithmetic, and formatted output. Policy values are constants near the top of the program.

```
# (name, price, data_GB, voice_minutes, validity_days,
#  excess_per_GB, excess_per_minute)
plans = [
    ("Starter", 900, 5, 300, 30, 120, 2),
    ("Flex", 1350, 12, 600, 30, 90, 1.5),
```

```
    ("Max", 1800, 25, 1200, 60, 70, 1),
    ("ShortPass", 500, 3, 150, 7, 150, 2),
]
usage_data = 10
usage_minutes = 450
required_days = 30
minimum_data = 8
minimum_minutes = 300
budget = 1600
candidates = []

for plan in plans:
    name, price, data, minutes, validity, data_rate, minute_rate = plan
    if price < 0 or data < 0 or minutes < 0 or validity < 0:
        print("Invalid plan:", name)
        continue
    excess_data = max(0, usage_data - data)
    excess_minutes = max(0, usage_minutes - minutes)
    total = price + excess_data * data_rate + excess_minutes * minute_rate
    eligible = (validity >= required_days and data >= minimum_data
                and minutes >= minimum_minutes and total <= budget)
    print(name, "excess data:", excess_data, "excess minutes:",
          excess_minutes, "estimated cost:", round(total, 2),
          "eligible:", eligible)
    if eligible:
        candidates.append((total, -data, name, excess_data, excess_minutes))

if len(candidates) == 0:
    print("No plan satisfies all usage, validity, and budget constraints.")
else:
    candidates.sort()
    best = candidates[0]
    print("Recommended plan:", best[2])
    print("Estimated total:", round(best[0], 2))
```

4. Sample run guidance

Run the program as written to see a sample result. Replace the sample lists and constants with the values provided in your assignment. Keep the rule order and units consistent; for example, percentages should be entered as percentages and converted to fractions only in the calculation.